

What is Cloud Firestore?

Cloud Firestore is a NoSQL cloud database provided by Firebase that:

- Store data in collection and documents
- Automatically syncs data in real-time across all connected devices
- Each document is structured like a JSON Object with fields
- Works offline and syncs when connection is restored
- Scales automatically when app grows

What are Collection and Documents

- A container (folder) that holds documents
- Like a category or table in trad... database
- Can have many documents inside.

Documents:

- A single record in a collection
- Like one row in a spreadsheet
- contains fields (key-value pairs)
- Has a unique ID (auto-generated)

How to add new document,

```
import { addDoc, collection, firestore } from  
  "firebase/firestore";
```

```
await addDoc(collection(db, "users"), {
```

```
  name: 'Shumaila',
```

```
  age: 20,
```

```
  email: 'shumaila@gmail.com',
```

```
});
```

How to read all data from a collection

```
const querySnapshot = await getDocs
```

```
(collection(db, "users"));
```

```
querySnapshot.forEach((doc) => {  
  console.log(doc.id, "=>", doc.data());  
});
```


What function is used to update a document?

→ Use `updateDoc()` along with `doc()`

```
const userRef = doc(db, 'users', 'docID');  
await updateDoc(userRef, {  
  age: 25,  
  email: 'newUser@gmail.com'  
});
```

How can you delete specific doc?

```
await deleteDoc(doc(db, 'users', 'docID'));
```

Advantages Firestore over traditional database

No Backend Required - No need to setup ^{node.js} server

Real-time sync - change appear instantly on all ^{small devices}

Offline Support - App works without internet

Why do we need `asyn/await` with Firestore

Firestore operation involves internet request to Google's Server, which take time to wait for these operation to

What is doc() reference?

Document Reference is like a pointer or address to a specific document in Firestore.

```
const docRef = doc('db', 'collectionName',  
                    "documentID");
```

- Point to a specific doc
- Don't fetch the data yet.
- Used for update, delete operations